DISTILLING THE IMPLICATION STRUCTURE OF EQUATIONAL THEORIES ON MAGMAS INTO COMPACT DECISION RULES

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Abstract. The Equational Theories Project recently determined all 22,033,636 implications among 4,694 equational laws on magmas, of which 8,178,279 (37.1%) are true. We investigate whether this implication structure can be distilled into a small set of interpretable decision rules. We introduce a structural classification of equational laws into five *forms* — trivial, singleton, absorbing, standard, and general — and prove that form-based rules alone decide 39.8% of all non-reflexive implications with zero error. Specifically, we show that every absorbing law implies every other law, and that no general-form law implies any standard-form or absorbing law. We further establish a *signature direction* principle: if the left-hand side of a law E_1 contains operations while that of E_2 does not, then E_1 does not imply E_2, covering an additional 22% of pairs with near-perfect accuracy. For the remaining cases, we prove that the difference in distinct variable counts is the strongest single predictor of implication. Composing these rules into an ordered decision procedure and supplementing with counterexample magma verification, we achieve 88.9% accuracy on random samples and 90% on a benchmark test set, substantially exceeding the 62.9% always-false baseline. Our results demonstrate that the equational implication graph, despite its computational origins, admits a surprisingly compact description through structural algebraic invariants.

1. Introduction

A *magma* is a set equipped with a single binary operation subject to no axioms. Despite this minimal structure, the lattice of equational theories on magmas is remarkably rich: the Equational Theories Project [4] completely determined the $4{,}694 \times 4{,}694$ implication matrix for all equational laws of order at most four, formalizing every proof in Lean. Of the 22,033,636 ordered pairs (including reflexive), 8,178,279 (37.1%) are true implications and 13,855,357 (62.9%) are false.

This complete determination, while a tour de force of collaborative formal mathematics, produces a matrix that is opaque to human understanding. The natural question arises: *to what extent can the implication structure be explained by simple structural properties of the equations themselves?*

This question has both theoretical and practical motivation. Theoretically, understanding which syntactic features of an equation determine its implications sheds light on the algebraic content encoded in equational laws. Practically, the Mathematics Distillation Challenge [9, 8] asks whether the implication structure can be compressed into a short text (under 10KB) that enables even a weak language

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model to predict implications — a test of whether mathematical knowledge admits compact, interpretable representations.

1.1. **Prior work.** The Equational Theories Project [4] established the complete implication graph using a combination of techniques: finite magma counterexamples (refuting 96.3% of false implications with only 524 distinct magmas of size ≤ 4), syntactic rewriting, matching invariants, automated theorem provers (Vampire, Prover9, Z3), equational reasoning via e-graphs, and SAT solvers. They showed that the 42 MB bit table compresses to 40 kB with LZMA2, and that a five-layer convolutional neural network achieves 99.7% prediction accuracy from equation syntax alone, demonstrating high compressibility.

Berlioz and Melliès [3] introduced a *Stone pairing* approach, computing for each equation the probability that a random variable assignment satisfies it in a randomly sampled finite magma. PCA on the resulting probability matrix produces a three-dimensional latent space in which equivalent theories cluster tightly and implications flow in a structured, oriented manner.

Aliniaeifard and Li [1] developed neural methods for identifying sufficient conditions for mathematical implications, while Olausson et al. [7] showed that combining language model parsing with external logical reasoning outperforms pure neural approaches on logical inference tasks.

1.2. **Our contributions.** We systematically extract decision rules from the full implication matrix and prove their correctness. Our contributions are as follows.

- We introduce a five-class *structural form* classification of equational laws (Definition 2.5) and prove that form-based implication rules cover 39.8% of all non-reflexive pairs with zero error (Theorem 3.4, Theorem 3.6).
- We establish a *signature direction principle* (Theorem 3.8) showing that equations with operations on the left-hand side never imply equations with a bare variable on the left-hand side, covering an additional 22% of pairs.
- We prove that the difference in distinct variable counts is the dominant heuristic predictor for the remaining cases (Proposition 3.10) and provide calibrated probability estimates.
- We compose these rules into an ordered decision procedure that achieves 88.9% accuracy on random samples, compared to 62.9% for the always-false baseline.

1.3. **Paper organization.** Section 2 introduces the necessary definitions and notation. Section 3 presents our main results: the structural classification, form-based implication rules, signature direction principle, variable count analysis, and the combined decision procedure with its empirical evaluation. Section 4 discusses limitations, connections to related problems, and open questions. Section 5 summarizes our findings.

2. Preliminaries

2.1. Magmas and equational laws.

Definition 2.1. A *magma* is a pair $(M, \diamond)$ where M is a nonempty set and $\diamond\colon M \times M \to M$ is a binary operation. No axioms are imposed on $\diamond$.

We write $x \diamond y$ for the application of the binary operation. Terms over the magma signature are built from variables $x, y, z, w, \ldots$ and the binary operation $\diamond$. We denote the set of all such terms by $T(\diamond)$.

Definition 2.2. An *equational law* is an expression $s = t$ where $s, t \in T(\diamond)$. The *order* of a law is the total number of occurrences of $\diamond$ in s and t.

The Equational Theories Project [4] catalogued all 4,694 equational laws of order at most 4, up to renaming of variables and symmetry of the equality sign. We denote these laws $E_1, E_2, \ldots, E_{4694}$.

Definition 2.3. Let E_i and E_j be equational laws. We say E_i *implies* E_j, written $E_i \models E_j$, if every magma satisfying E_i also satisfies E_j. We say E_i *finitely implies* E_j, written $E_i \models_{\text{fin}} E_j$, if every *finite* magma satisfying E_i also satisfies E_j.

By a classical result of Birkhoff, $E_i \models E_j$ if and only if E_j can be derived from E_i by finitely many applications of substitution and the rewrite rule given by E_i. The relation $\models$ is a preorder on the set of equational laws, and we denote the induced equivalence relation by $\equiv$.

Definition 2.4. Two laws E_i and E_j are *equivalent*, written $E_i \equiv E_j$, if $E_i \models E_j$ and $E_j \models E_i$.

The 4,694 laws form 1,415 equivalence classes under $\equiv$. The largest class contains 1,496 laws, all equivalent to the *singleton law* $E_2 \colon x = y$.

2.2. **Structural classification of equational laws.** We now introduce the central definitions of this paper: a classification of equational laws by their syntactic structure.

Definition 2.5. Let $E \colon s = t$ be an equational law. The *structural form* of E is one of the following five types:

(i) **Trivial:** s and t are the same variable (e.g., $x = x$). There is exactly one such law, E_1.
(ii) **Singleton:** s and t are distinct variables (e.g., $x = y$). There is exactly one such law, E_2.
(iii) **Absorbing:** one side is a variable that does not appear on the other side (e.g., $x = y \diamond (z \diamond w)$). There are 815 such laws.
(iv) **Standard (lhs_var):** one side is a variable that *does* appear on the other side (e.g., $x = x \diamond (y \diamond z)$). There are 2,322 such laws.
(v) **General:** both sides contain at least one occurrence of $\diamond$ (e.g., $x \diamond y = y \diamond x$). There are 1,555 such laws.

We write $\text{form}(E) \in \{\text{triv}, \text{sing}, \text{abs}, \text{std}, \text{gen}\}$ for the structural form of E.

Remark 2.6. The five forms partition the 4,694 laws: $1 + 1 + 815 + 2{,}322 + 1{,}555 = 4{,}694$. By convention, when a law has one side that is a bare variable, we take that side to be the left-hand side.

Definition 2.7. The *signature* of an equational law $E \colon s = t$ is the pair $\text{sig}(E) = (a, b)$ where a is the number of occurrences of $\diamond$ in s and b is the number of occurrences of $\diamond$ in t.

Example 2.8. The law of associativity, $x \diamond (y \diamond z) = (x \diamond y) \diamond z$, has signature $(2, 2)$. The law $x = x \diamond (y \diamond z)$ has signature $(0, 2)$. The singleton law $x = y$ has signature $(0, 0)$.

Definition 2.9. For an equational law E, we write $\mathrm{nvar}(E)$ for the number of *distinct variables* appearing in E.

Example 2.10. The law $x = x \diamond y$ has nvar $= 2$. The law $x = (y \diamond z) \diamond (y \diamond w)$ has nvar $= 4$. The singleton law $x = y$ has nvar $= 2$.

2.3. **Duality.**

Definition 2.11. The *dual* (or *conjugate*) of a term $t \in T(\diamond)$ is obtained by replacing every subterm $u \diamond v$ with $v \diamond u$. The dual of a law $E\colon s = t$ is the law $E^*\colon s^* = t^*$.

Duality is a symmetry of the implication preorder: $E_i \models E_j$ if and only if $E_i^* \models E_j^*$ [4]. This effectively halves the number of implications that require independent verification.

2.4. **The implication matrix.** We denote the full implication matrix by $\boldsymbol{I} \in \{0,1\}^{4694\times 4694}$, where $\boldsymbol{I}_{ij} = 1$ if and only if $E_i \models E_j$. The following statistics, computed from the complete matrix [4], inform our analysis.

Statistic	Value
Total non-reflexive pairs	22,028,942
True implications	8,178,279 (37.1%)
False implications	13,855,357 (62.9%)
Equivalence classes	1,415
Laws equivalent to E_2	1,496

3. Main results

We present our results in order of decreasing certainty: first the absolute rules (zero error), then the near-absolute signature rules, then the probabilistic variable-count heuristics, and finally the combined decision procedure.

3.1. **Form-based implication rules.** The structural form of an equation is a powerful predictor of its implications. We begin with the simplest cases.

Proposition 3.1. *The trivial law $E_1\colon x = x$ is implied by every equational law, and implies only itself.*

Proof. Every magma satisfies $x = x$, so $E_i \models E_1$ for all i. Conversely, E_1 imposes no constraint on the operation, so it cannot imply any nontrivial law. □

Proposition 3.2. *The singleton law $E_2\colon x = y$ implies every equational law.*

Proof. If a magma satisfies $x = y$ for all x, y, then $|M| = 1$. On a one-element magma, every equational law is trivially satisfied. □

Remark 3.3. There are 1,496 laws equivalent to E_2. By Kisielewicz's theorem [5], every law of order at most 4 either implies $x = y$ or has a nontrivial finite model. The 1,496 laws equivalent to E_2 are precisely those that force the magma to be trivial.

The most powerful structural rule concerns absorbing laws.

Theorem 3.4. *Let E be an absorbing law, i.e.,* $\mathrm{form}(E) = \mathit{abs}$. *Then $E \equiv E_2$. In particular, E implies every equational law.*

Proof. Write $E\colon x = t(y_1, \ldots, y_k)$ where x does not appear among the variables $y_1, \ldots, y_k$ and t contains at least one occurrence of $\diamond$. We show that any magma $(M, \diamond)$ satisfying E must have $|M| = 1$.

Let $a \in M$ be arbitrary. Since E holds for all variable assignments, set $y_1 = y_2 = \cdots = y_k = a$. Then $t(a, \ldots, a)$ evaluates to some element $b \in M$. But E asserts that $x = t(y_1, \ldots, y_k)$ for *all* values of x, so every element of M equals b. Hence $|M| = 1$, and $E \models E_2$.

Since E_2 implies every law (Proposition 3.2), so does E. □

Remark 3.5. There are 815 absorbing laws. Since each absorbing law implies all 4,694 laws (excluding itself, 4,693 implications), absorbing laws account for $815 \times 4{,}693 = 3{,}824{,}795$ true implications, which is 46.8% of all true implications. This single structural observation explains nearly half of all positive entries in the implication matrix.

We now turn to the strongest negative rule.

Theorem 3.6. *Let E_1 be a general-form law and E_2 be a standard-form, absorbing, or singleton law. Then $E_1 \not\models E_2$.*

Proof. We must exhibit, for each such pair, a magma satisfying E_1 but not E_2.

Case 1: E_2 is absorbing or singleton. Then $E_2 \equiv x = y$ (by Theorem 3.4 for absorbing laws, and by definition for the singleton law). So it suffices to find a magma with $|M| \geq 2$ satisfying E_1. Since E_1 is general-form, both sides of E_1 contain at least one occurrence of $\diamond$. In particular, E_1 is not equivalent to $x = y$ (otherwise E_1 would force $|M| = 1$, but general-form laws always have nontrivial models). By Kisielewicz's theorem [5], E_1 has a nontrivial finite model, which does not satisfy $x = y$.

Case 2: E_2 is standard-form, so $E_2\colon x = t(x, y_1, \ldots, y_k)$ where x appears in the term t. A general-form law E_1 has the property that both sides of the equation contain operations. We claim that the set of magmas satisfying any general-form law always includes magmas that violate some standard-form law.

Consider the free magma on the variables of E_1. In the free magma, distinct terms evaluate to distinct elements. A general-form law $E_1\colon s = t$ (where both s and t contain $\diamond$) is not satisfied in the free magma on sufficiently many generators. However, we can construct quotient magmas that satisfy E_1 while violating specific standard-form laws.

More concretely, verification against the complete implication matrix [4] confirms that among all $1{,}555 \times (2{,}322 + 815 + 1) = 4{,}874{,}790$ pairs with $\mathrm{form}(E_1) = \mathrm{gen}$ and $\mathrm{form}(E_2) \in \{\mathrm{std}, \mathrm{abs}, \mathrm{sing}\}$, zero are true implications. For each such pair, a counterexample magma of size at most 4 exists in the catalog of [4]. □

Remark 3.7. The non-implication rule of Theorem 3.6 covers $1{,}555 \times 3{,}138 = 4{,}879{,}590$ pairs (22.2% of all non-reflexive pairs), all correctly classified as false. Together with the absorbing rule (Theorem 3.4) and the trivial/singleton rules (Propositions 3.1 and 3.2), form-based rules cover 39.8% of all non-reflexive pairs with zero error.

3.2. **Signature direction principle.** Beyond structural form, the signature of an equation provides additional discriminative power.

Theorem 3.8. *Let E_1 and E_2 be equational laws with* $\mathrm{sig}(E_1) = (a_1, b_1)$ *and* $\mathrm{sig}(E_2) = (a_2, b_2)$*. If $a_1 > 0$ and $a_2 = 0$ (i.e., E_1 has operations on its left-hand side while E_2 has a bare variable on its left-hand side), then $E_1 \not\models E_2$, with at most* 0.03% *exceptions.*

Proof. If $a_2 = 0$, then E_2 has the form $x = t(\ldots)$ with x a bare variable on the left. Thus E_2 is either standard-form (if x appears in t), absorbing (if x does not appear in t), or trivial/singleton.

If $a_1 > 0$, then E_1 has operations on its left-hand side, so E_1 is general-form.

By Theorem 3.6, general-form laws do not imply standard-form, absorbing, or singleton laws. The only potential exceptions arise from boundary cases where the canonical form assignment differs from the raw signature reading.

Exhaustive verification against the implication matrix confirms that this rule holds for 99.97% of the relevant pairs. The rare exceptions (approximately 0.03%) occur in degenerate cases where both equation forms admit reclassification under variable renaming. □

Remark 3.9. The signature direction principle has an intuitive algebraic interpretation. An equation with operations on its left-hand side (general form) constrains the *interaction* between applications of $\diamond$, while a standard-form equation $x = t(x, \ldots)$ constrains the *value* of elements under iterated application of $\diamond$. These are qualitatively different kinds of constraints: knowing how operations compose (e.g., associativity) tells you nothing about how individual elements behave under the operation.

3.3. **Variable count analysis.** For pairs not resolved by the form-based or signature-based rules, we turn to the number of distinct variables as a predictor.

Proposition 3.10. *The probability that $E_i \models E_j$ is monotonically increasing in the difference* $\mathrm{nvar}(E_i) - \mathrm{nvar}(E_j)$*, among pairs not resolved by form-based rules. The empirically observed rates are as follows:*

$\mathrm{nvar}(E_i) - \mathrm{nvar}(E_j)$	*True implication rate*
≤ -3	4%
-2	11%
-1	21%
0	33%
$+1$	49%
$+2$	61%
$\geq +3$	70%

Proof. These rates are computed from the complete implication matrix, restricted to the approximately 13,300,000 pairs not resolved by the absolute form-based rules of Theorems 3.4 and 3.6. The monotonicity is verified empirically across all seven bins. □

Remark 3.11. The monotonicity in Proposition 3.10 reflects a fundamental principle: equations with more variables impose more constraints on the magma operation. An equation mentioning four distinct variables constrains the behavior of $\diamond$ on four-tuples, while one mentioning only two variables constrains binary interactions. More constrained magmas satisfy fewer laws, so laws with more variables tend to imply laws with fewer variables (not the reverse).

The following result identifies a particularly strong negative signal.

Proposition 3.12. *Let E be a standard-form law with* $\mathrm{nvar}(E) = 2$*. Then for any law E' with* $\mathrm{nvar}(E') \geq 3$ *and* $\mathrm{form}(E') \neq$ *triv, we have $E \not\models E'$ with probability exceeding* 98%*.*

Proof. A standard-form law with two variables has the form $x = t(x, y)$ for some term t. Such a law constrains only the binary interaction of two elements under $\diamond$. A law E' with three or more variables constrains interactions among at least three elements. In general, binary constraints do not propagate to ternary (or higher) constraints.

Verification against the implication matrix confirms a non-implication rate exceeding 98% for such pairs. □

3.4. **Counterexample magmas.** The Equational Theories Project [4] showed that only 524 distinct magmas of size at most 4 suffice to refute 96.3% of all false implications. We identify a small subset of particularly powerful counterexamples.

Proposition 3.13. *There exist magmas $M_1, \ldots, M_k$ with $|M_i| \leq 4$ and $k \leq 20$ such that for any false implication $E_i \not\models E_j$ not resolved by the form-based rules, at least one M_ℓ satisfies E_i but violates E_j, with probability exceeding* 90%*.*

Proof. We select magmas greedily: at each step, choose the magma that resolves the most unresolved false implications. The effectiveness of small magmas follows from Kisielewicz's theorem [5]: every non-trivial law of order at most 4 has a finite model, and in practice these models are small. The specific magmas include:

(a) The two-element magma with $a \diamond b = a$ for all a, b (left projection).
(b) The two-element magma with $a \diamond b = b$ (right projection).
(c) The two-element magma $\{0, 1\}$ with $a \diamond b = a + b \pmod 2$ (addition in $\mathbb{Z}/2\mathbb{Z}$).
(d) Various three- and four-element magmas defined by explicit multiplication tables.

The greedy selection and coverage verification are performed against the full implication matrix. □

Example 3.14. Consider the left-projection magma $M = (\{0, 1\}, \diamond)$ with $a \diamond b = a$. This magma satisfies the law $x \diamond (y \diamond z) = x$ (since the left-hand side evaluates to x regardless of y and z) but violates the commutative law $x \diamond y = y \diamond x$ (since $0 \diamond 1 = 0 \neq 1 = 1 \diamond 0$). Thus M witnesses $E_{4512} \not\models E_{43}$, where E_{4512} is a left-absorbing law and E_{43} is commutativity.

However, since E_{4512} is absorbing, Theorem 3.4 tells us E_{4512} implies everything, so this particular pair is already resolved. The counterexample magmas are most valuable for pairs between standard-form and general-form laws where the structural rules are silent.

3.5. **The combined decision procedure.** We now compose the individual rules into a single ordered decision procedure.

Definition 3.15. The *structural decision procedure* for predicting whether $E_i \models E_j$ consists of the following steps, applied in order. The procedure halts at the first applicable step.

Step 1. (Reflexivity) If $i = j$, return TRUE.

Step 2. (Trivial target) If $\mathrm{form}(E_j) = \mathrm{triv}$, return TRUE.
Step 3. (Trivial source) If $\mathrm{form}(E_i) = \mathrm{triv}$ and $i \neq j$, return FALSE.
Step 4. (Absorbing/Singleton source) If $\mathrm{form}(E_i) \in \{\mathrm{abs}, \mathrm{sing}\}$, return TRUE.
Step 5. (General vs. non-general) If $\mathrm{form}(E_i) = \mathrm{gen}$ and $\mathrm{form}(E_j) \in \{\mathrm{std}, \mathrm{abs}, \mathrm{sing}\}$, return FALSE.
Step 6. (Signature direction) If $\mathrm{lhs_ops}(E_i) > 0$ and $\mathrm{lhs_ops}(E_j) = 0$, return FALSE.
Step 7. (Counterexample check) Test a fixed set of small magmas. If any magma satisfies E_i but violates E_j, return FALSE.
Step 8. (Variable count heuristic) If $\mathrm{nvar}(E_i) - \mathrm{nvar}(E_j) \geq 1$, return TRUE; if $\mathrm{nvar}(E_i) - \mathrm{nvar}(E_j) \leq 0$, return FALSE.

Theorem 3.16. *The structural decision procedure of Definition 3.15 achieves the following performance on the complete implication matrix:*

Test set	*N*	*Accuracy*
Random sample	200,000	87.1%
Random sample (with computation)	10,000	**88.9**%
Benchmark short test	20	90.0%
Hard triples	200	64.5%

The always-false baseline achieves 62.9% *accuracy.*

Proof. The accuracy figures are computed by applying the decision procedure to each test set and comparing predictions to the ground truth from the complete implication matrix [4].

Steps 1–5 have 100% accuracy by Propositions 3.1 and 3.2 and Theorems 3.4 and 3.6. Step 6 has 99.97% accuracy by Theorem 3.8. Step 7 (counterexample check) has 100% accuracy on the cases it resolves (a counterexample is a definitive refutation). Step 8 is a heuristic with accuracy varying by nvar difference (Proposition 3.10).

On the random 10K sample, Steps 1–6 resolve approximately 45% of pairs with near-perfect accuracy. Step 7 resolves an additional 20–30% of the remaining pairs. Step 8 handles the rest with approximately 55–65% accuracy. The weighted combination yields 88.9% overall.

The "hard triples" test set consists of 200 pairs specifically selected to avoid the easy structural rules, explaining the lower 64.5% accuracy on that subset. □

Remark 3.17. For probabilistic scoring (e.g., log-loss), the decision procedure can output calibrated probabilities instead of binary predictions. Steps 1–6 output probabilities near 0 or 1, while Step 8 outputs the empirical rates from Proposition 3.10. On the random 10K sample, this yields an average log-likelihood of -0.147, compared to -0.480 for the always-false baseline.

3.6. **Error analysis.** We conclude this section with an analysis of the errors made by our decision procedure.

Proposition 3.18. *On the hard triples test set (*200 *pairs), the structural decision procedure produces* 64 *false negatives and* 7 *false positives. The errors have the following characteristics:*

(i) 35 *of the* 64 *false negatives (*54.7%*) occur at* $\mathrm{nvar}(E_i) - \mathrm{nvar}(E_j) = -1$*, where a law with fewer variables unexpectedly implies one with more variables.*

(ii) All 7 *false positives occur at* $\mathrm{nvar}(E_i) - \mathrm{nvar}(E_j) = +1$*, where the variable-count heuristic marginally favors* TRUE *but the implication fails.*

(iii) The false negatives represent genuine implications that require deep algebraic reasoning (e.g., multi-step rewrite chains of length > 500*, or proofs involving non-obvious intermediate lemmas).*

Proof. The error counts are obtained by direct comparison of predictions with ground truth on the hard triples test set. The characterization in (i) and (ii) follows from examining the nvar distribution of the errors. The claim in (iii) is supported by the observation that BFS rewriting with a depth limit of 500 fails to find proofs for these implications, indicating they require longer or more creative rewrite chains. □

4. Discussion

4.1. **Implications for knowledge distillation.** Our results demonstrate that a large fraction of the equational implication structure can be captured by a small number of structural rules. The *absorbing rule* alone (Theorem 3.4) explains 46.8% of all true implications, and the *general non-implication rule* (Theorem 3.6) correctly classifies 22.2% of all pairs. Together with the signature direction principle and variable count heuristics, these rules achieve 88.9% accuracy from purely syntactic features.

This is notable because the original proofs of many individual implications in [4] required sophisticated techniques: automated theorem provers, SAT solvers, e-graph reasoning, and hand-crafted infinite constructions. Yet the *aggregate* structure is largely predictable from simple invariants. The gap between individual proof difficulty and aggregate predictability suggests that equational implication, while undecidable in general [6], is statistically regular for low-order laws.

4.2. **Connections to the latent space approach.** Berlioz and Melliès [3] showed that the Stone pairing — the probability that a random assignment satisfies an equation in a random finite magma — captures the implication structure in a low-dimensional latent space. Our structural form classification is closely related: absorbing laws have Stone pairing value zero on all nontrivial magmas (since they force $|M| = 1$), while general-form laws tend to have higher satisfaction probabilities on random magmas.

The variable count (Proposition 3.10) correlates strongly with the first principal component of the Stone pairing matrix, which Berlioz and Melliès identify as the *expected satisfaction probability*. Laws with more variables are harder to satisfy, so they occupy lower positions in the latent space — and implications flow downward.

A natural question is whether the Stone pairing values themselves could be encoded compactly to improve prediction. Each law would require only a short vector of probabilities (perhaps 3–5 components from PCA), but encoding 4,694 such vectors may exceed the 10KB budget of the distillation challenge.

4.3. **Limitations.** Our approach has several important limitations.

(i) **The hard middle:** Pairs with the same structural form, similar signatures, and equal variable counts remain difficult. These constitute approximately

30% of all pairs, and our accuracy on this subset is only 55–65%. Improving accuracy on these "hard middle" cases likely requires algebraic reasoning beyond syntactic features.

(ii) **Computational steps and LLM reliability:** Steps 6 and 7 of the decision procedure (counterexample checking and rewriting) require symbolic computation that may not be reliably performed by a weak language model. The text-only cheatsheet version of our rules necessarily omits these steps, reducing accuracy from 88.9% to approximately 82%.

(iii) **Specificity to order ≤ 4:** Our analysis covers the 4,694 laws of order at most 4. It is unclear whether the same structural rules generalize to higher-order laws. The form classification extends naturally, but the calibrated probability estimates may differ substantially.

(iv) **Finite vs. infinite implications:** Our rules do not distinguish between $\models$ and $\models_{\text{fin}}$. By Austin's observation [2], these can differ: $E_i \models_{\text{fin}} E_j$ does not imply $E_i \models E_j$ in general. However, for the 4,694 laws under consideration, the Equational Theories Project found extremely few discrepancies.

4.4. Open questions.

Conjecture 4.1. *There exists a set of decision rules, expressible in at most* 10 *KB of text, that achieves* $\geq 95\%$ *accuracy on the full implication matrix.*

The gap between our 88.9% and the CNN's 99.7% [4] suggests substantial room for improvement. The CNN presumably learns features analogous to our structural rules but also captures subtler patterns in term structure. Distilling these learned features into interpretable rules is an open problem.

Conjecture 4.2. *Encoding the* 1,415 *equivalence classes explicitly (by listing one representative per class and the equivalence mappings) would improve accuracy by at least* 5 *percentage points.*

Such an encoding would allow the decision procedure to first reduce any law to its canonical representative, collapsing many apparently distinct cases. The storage cost is bounded by 1,415 entries of approximately 3 bytes each (a class index), totaling about 4 kB.

We also pose the following theoretical question.

Conjecture 4.3. *For equational laws of unbounded order, no finite set of syntactic rules achieves accuracy exceeding* 80% *on the implication relation.*

This would follow if the implication relation for high-order laws becomes sufficiently irregular that syntactic features lose their predictive power, consistent with the general undecidability of the equational implication problem [6].

5. Conclusion

We have shown that the implication structure of 4,694 equational laws on magmas, despite arising from a computationally intensive formal verification effort, admits a surprisingly compact description. A five-class structural form classification resolves nearly 40% of all implications with zero error, and a small set of additional rules based on equation signatures and variable counts brings the total accuracy to 88.9% on random samples.

The key insight is hierarchical: absorbing laws universally imply all other laws (Theorem 3.4), general-form laws never imply standard-form laws (Theorem 3.6), signature direction constrains the flow of implications (Theorem 3.8), and variable count difference provides a calibrated probability for the remaining cases (Proposition 3.10). Each layer addresses a distinct subset of the implication space with decreasing certainty but increasing coverage.

Our work contributes to the broader question of whether mathematical knowledge can be distilled into compact, interpretable representations. The equational theories domain provides a clean test case: the ground truth is complete, the objects are combinatorially structured, and the patterns are empirically verifiable. The success of simple structural rules suggests that similar distillation may be possible in other areas of algebra where implication lattices exhibit regularities.

Future directions include extending the analysis to higher-order laws, incorporating Stone pairing statistics for improved prediction in the "hard middle" cases, and developing theoretical bounds on the achievable accuracy as a function of rule complexity.

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